

In [9], Caro, Lev, Roditty, Tuza and Yuster investigated the graphs with small rainbow connection numbers, and they gave a sufficient condition that guarantees $rc(G) = 2$.

Theorem 2.44 [9] *Any non-complete graph with $\delta(G) \geq n/2 + \log n$ has $rc(G) = 2$.*

We know that having diameter 2 is a necessary requirement for having $rc(G) = 2$, although certainly not sufficient (e.g., consider a star). Clearly, if $\delta(G) \geq n/2$ then $\text{diam}(G) = 2$, but we do not know if this guarantees $rc(G) = 2$. The above theorem shows that by slightly increasing the minimum degree assumption, $rc(G) = 2$ follows.

Kemnitz and Schiermeyer [34] gave a sufficient condition to guarantee $rc(G) = 2$ according to the number of edges m .

Theorem 2.45 [34] *Let G be a connected graph of order n and size m . If $\binom{n-1}{2} + 1 \leq m \leq \binom{n}{2} - 1$, then $rc(G) = 2$.*

Let $G = G(n, p)$ denote, as usual [3], the random graph with n vertices and edge probability p . For a graph property A and for a function $p = p(n)$, we say that $G(n, p)$ satisfies A *almost surely* if the probability that $G(n, p(n))$ satisfies A tends to 1 as n tends to infinity. We say that a function $f(n)$ is a *sharp threshold function* for the property A if there are two positive constants c and C so that $G(n, cf(n))$ almost surely does not satisfy A and $G(n, p)$ satisfies A almost surely for all $p \geq Cf(n)$. It is well known that all monotone graph properties have a sharp threshold function (see [6] and [27]). Since having $rc(G) \leq 2$ is a monotone graph property (adding edges does not destroy this property), it has a sharp threshold function. The following theorem establishes it.

Theorem 2.46 [9] *$p = \sqrt{\log n/n}$ is a sharp threshold function for the graph property $rc(G(n, p)) \leq 2$.*

Theorem 2.44 asserts that minimum degree $n/2 + \log n$ guarantees $rc(G) = 2$. Clearly, minimum degree $n/2 - 1$ does not, as there are connected graphs with minimum degree $n/2 - 1$ and diameter 3 (just take two vertex-disjoint cliques of order $n/2$ each and connect them by a single edge. It is therefore interesting to raise:

Problem 2.47 [9] *Determine the minimum degree threshold that guarantees $rc(G) = 2$.*

By Proposition 2.1, we know that the problem of considering graphs with $rc(G) = 2$ is equivalent to that of considering graphs with $\text{src}(G) = 2$.

A bipartite graph which is not complete has diameter at least 3. A proof similar to that of Theorem 2.46 gives the following result.

Theorem 2.48 [9] *Let $c = 1/\log(9/7)$. If G is a non-complete bipartite graph with n vertices and any two vertices in the same vertex class have at least $2c\log n$ common neighbors in the other vertex class, then $rc(G) = 3$.*

The following theorem asserts however that having diameter 2 and only logarithmic minimum degree suffices to guarantee rainbow connection 3.

Theorem 2.49 [10] *If G is an n -vertex graph with diameter 2 and minimum degree at least $8\log n$, then $rc(G) \leq 3$.*

Since a graph with minimum degree $n/2$ is connected and has diameter 2, we have as an immediate result [10]: If G is an n -vertex graph with minimum degree at least $n/2$ then $rc(G) \leq 3$. We know that any graph G with $rc(G) = 2$ must have $diam(G) = 2$, so graphs with $rc(G) = 2$ belong to the graph class with $diam(G) = 2$. By Corollary 2.28, we know that for a bridgeless graph with $diam(G) = 2$, $rc(G) \leq r(r+2) \leq 8$. As $rc(G)$ is at least the number of bridges of G , so it may be very large if the number of bridges of G is sufficiently large by Observation 2.11. So there is an interesting problem:

Problem 2.50 *For any bridgeless graph G with $diam(G) = 2$, determine the smallest constant c such that $rc(G) \leq c$.*

Recently, Li, Li and Liu [39] derived that $c \leq 5$ by showing the following result:

Theorem 2.51 [39] *$rc(G) \leq 5$ if G is a bridgeless graph with diameter 2; and that $rc(G) \leq k + 2$ if G is a connected graph with diameter 2 and k bridges, where $k \geq 1$.*

In the proof, Li, Li and Liu derived that if G is a bridgeless graph with order n and diameter 2, then it is either 2-connected, or it has only one cut vertex v , furthermore, v is the center of G with radius 1. They showed that 5 is almost best possible as there are infinity many bridgeless graphs with diameter 2 whose rainbow connection numbers are 4, however they have not found examples of such graphs with $rc(G) = 5$. The bound $k + 2$ is sharp as there are infinity graphs with diameter 2 and k bridges whose rainbow connection numbers attain this bound [39].

In [46] and [47], Li and Sun investigated the graphs with large rainbow connection numbers and strong rainbow connection numbers, respectively. They derived the two following results. Note that each path P_j in the member of graph class $\mathcal{G}_i$ ($1 \leq i \leq 4$) of Figure 2.3 may be trivial.

Theorem 2.52 [46] *For a connected graph G with m edges, we have $rc(G) \neq m - 1$; and $rc(G) = m - 2$ if and only if G is a 5-cycle or belongs to one of four graph classes $\mathcal{G}_i$'s ($1 \leq i \leq 4$) shown in Figure 2.3.*

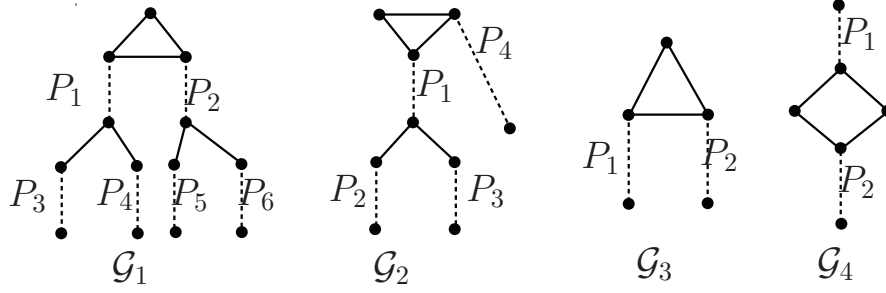


Figure 2.3 The figure for the four graph classes.

We now introduce two graph classes. Let C be the cycle of a unicyclic graph G , $V(C) = \{v_1, \dots, v_k\}$ and $\mathcal{T}_G = \{T_i : 1 \leq i \leq k\}$ where T_i is the unique tree containing vertex v_i in subgraph $G \setminus E(C)$. We say T_i and T_j are *adjacent* (*nonadjacent*) if v_i and v_j are adjacent (nonadjacent) in cycle C . Then let

$\mathcal{G}_5 = \{G : G \text{ is a unicyclic graph, } k = 3, \mathcal{T}_G \text{ contains at most two nontrivial elements}\},$

$\mathcal{G}_6 = \{G : G \text{ is a unicyclic graph, } k = 4, \mathcal{T}_G \text{ contains two nonadjacent trivial elements and the other two (nonadjacent) elements are paths}\}.$

Theorem 2.53 [47] *For a connected graph G with m edges, $\text{src}(G) \neq m-1$; $\text{src}(G) = m-2$ if and only if G is a 5-cycle or belongs to one of the $\mathcal{G}_i$'s ($i = 5, 6$).*

By Proposition 2.1, Theorems 2.52 and 2.53, we investigated the graphs with $\text{rc}(G) \geq m-2$ ($\text{src}(G) \geq m-2$). Furthermore, we have the following interesting problem.

Problem 2.54 *Give a sufficient condition to guarantee $\text{rc}(G) \geq \alpha m$ ($\text{src}(G) \geq \alpha m$), where $0 < \alpha < 1$.*

2.5 For graph operations

Products of graphs occur naturally in discrete mathematics as tools in combinatorial constructions, they give rise to important classes of graphs and deep structural problems. The extensive literature on products that has evolved over the years presents a wealth of profound and beautiful results [32]. In [46], Li and Sun obtained some results on the rainbow connection numbers of products of graphs, including Cartesian product, composition (lexicographic product), union of graphs, etc. Actually, we know that the line graph of a graph is also a graph operation.

We first introduce the rainbow connection number of the Cartesian product of some graphs. The *Cartesian product* of graphs G and H is the graph $G \square H$ whose vertex set is $V(G) \times V(H)$ and whose edge set is the set of all pairs $(u_1, v_1)(u_2, v_2)$ such that either $u_1 u_2 \in E(G)$ and $v_1 = v_2$, or $v_1 v_2 \in E(H)$ and $u_1 = u_2$. The *strong product* of G and H is the graph $G \boxtimes H$ whose vertex set is $V(G) \times V(H)$ and whose edge set is the set of all pairs